

Topic 3 — Conservation of Energy

Edexcel GCSE (9-1) Physics (1PH0) — Paper 1

3.1 The Conservation Principle

Energy cannot be created or destroyed. It can only be **stored**, **transferred** between stores, or **dissipated** to the surroundings (usually as heat or sound).

Energy Stores and Pathways

Energy stores	Transfer pathways
Kinetic, gravitational, chemical, elastic, thermal (internal), nuclear, magnetic, electrostatic	Mechanical working, electrical working, heating, radiation

3.2 Kinetic Energy and GPE

$$KE = \frac{1}{2} \times m \times v^2$$

$$GPE = m \times g \times h$$

Worked Example: A 0.2 kg ball is dropped from 5 m ($g = 10$, no air resistance). Find the speed on landing.

$$GPE = 0.2 \times 10 \times 5 = 10 \text{ J} = KE = \frac{1}{2} \times 0.2 \times v^2 \rightarrow v^2 = 100 \rightarrow \mathbf{v = 10 \text{ m/s.}}$$

3.3 Power and Efficiency

$$P = E \div t \quad (\text{watt} = \text{joule/second})$$

$$\text{Efficiency} = \text{useful output} \div \text{total input} (\times 100\%)$$

Sankey Diagram — 60 W Bulb (10% efficient)



Arrow widths are proportional to the amounts of energy (or power).

3.4 Energy Resources

Renewable	Non-renewable
Solar, wind, hydroelectric, tidal, wave, geothermal, biomass	Coal, oil, natural gas, nuclear (uranium)

Comparison of Key Resources

Resource	Pros	Cons
Wind	Renewable, no CO ₂ in use, low running cost	Intermittent, visual/ecological impact
Solar	Renewable, no CO ₂ in use, modular	Intermittent (no power at night), large area

Nuclear	Reliable baseload, high energy density, low CO ₂	Radioactive waste, high upfront cost
Coal/gas	Reliable, cheap to build, proven technology	Emit CO ₂ , finite, air pollution

3.5 Reducing Unwanted Energy Transfers

- **Lubrication** reduces friction in moving parts.
- **Loft insulation** traps air to cut conduction/convection.
- **Cavity wall insulation** same principle for walls.
- **Double glazing** uses an air gap for windows.
- **Draught excluders** reduce unwanted airflow.

Exam Tip: The "best" reduction often depends on where most of the heat is lost. In a typical uninsulated UK house, walls lose the most heat (about a third) and the roof about a quarter, but loft insulation is cheap and quick to fit, so it is usually the first upgrade.

3.6 Specific Heat Capacity

$$E = m \times c \times \Delta T$$

c is in J/(kg °C). Water: $c \approx 4\,200$ J/(kg °C).

Worked Example: Heat 0.5 kg of water from 20 °C to 100 °C.

$$\Delta T = 80\text{ °C} \rightarrow E = 0.5 \times 4\,200 \times 80 = \mathbf{168\,000\text{ J}}.$$

Key Points Summary

- Energy is always conserved — never created or destroyed.
- $KE = \frac{1}{2}mv^2$; $GPE = mgh$; you should be able to rearrange both.
- Efficiency = useful/total ($\leq 100\%$).
- Reduce waste with insulation, double glazing and lubrication.
- Know the renewables/non-renewables list and pros/cons.